

Project Title: Customer Feedback Sentiment Analysis Dashboard

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GitHub Repository: <https://github.com/anshhh1101/Customer-Feedback-Sentiment-Analysis.git>

1. Problem Statement

In today's competitive business environment, organizations receive large volumes of customer feedback across multiple channels. However, manually analyzing this feedback is time-consuming, error-prone, and rarely fast enough to support real-time decision-making. Without an integrated system, businesses struggle to connect customer sentiment with actual sales and profit outcomes.

Key challenges addressed by this project:

- No clear visibility into customer sentiment — Positive, Neutral, or Negative — across products and regions
- Difficulty identifying loss-making products, sub-categories, and geographic regions
- Inability to visualize the relationship between discount strategies, sales volume, and profitability
- Lack of an integrated, interactive tool accessible to non-technical business users

2. Solution / Project Overview

To address these challenges, a two-page interactive Power BI Dashboard was designed and developed using the Sample Superstore retail dataset. The dashboard integrates customer sentiment classifications with core business KPIs — Sales, Profit, and Orders — enabling a comprehensive, filterable view of organizational performance without requiring any technical expertise from the end user.

Page 1: Customer Sentiment Overview

Provides a high-level business summary that updates dynamically based on the selected sentiment filter. Key components include KPI cards showing Total Sales (1.05M), Total Profit (300.18K), and Total Quantity (4,641); a Bing Map for geographic sales distribution; a scatter plot of Sales vs. Profit by Category; a regional breakdown table (South, Central, West, East); a pie chart for customer segments (Consumer, Corporate, Home Office); a funnel chart by sales stage (Awareness → Consideration → Purchase); and a sentiment-filtered bar chart for category-level sales performance.

Page 2: Profit & Loss Analysis

Provides deep profitability insights for management-level review. Key components include a horizontal bar chart comparing Sales vs. Profit by Region, a scatter plot mapping Discount vs. Profit by Category, a vertical bar chart showing Profit by Category (Technology: 145K, Office Supplies: 122K, Furniture: 18K), and a dedicated loss analysis chart identifying the top loss-making sub-categories: Tables (−17.7K), Bookcases (−3.5K), and Supplies (−1.2K).

3. Dashboard Screenshots

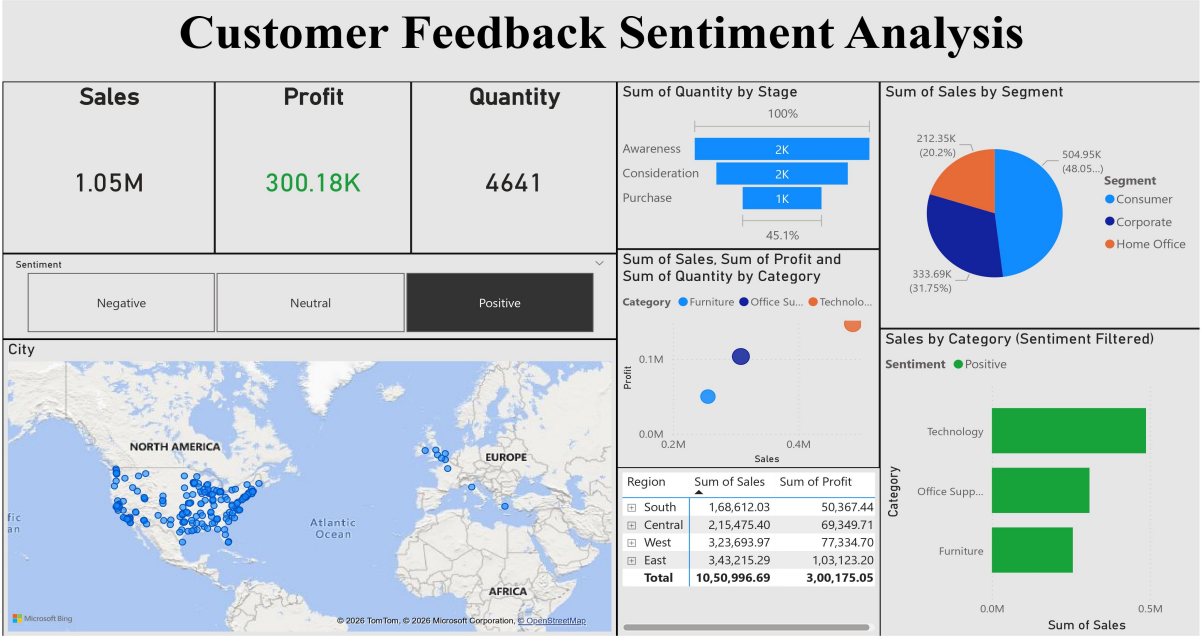


Figure 1: Customer Sentiment Overview — Page 1

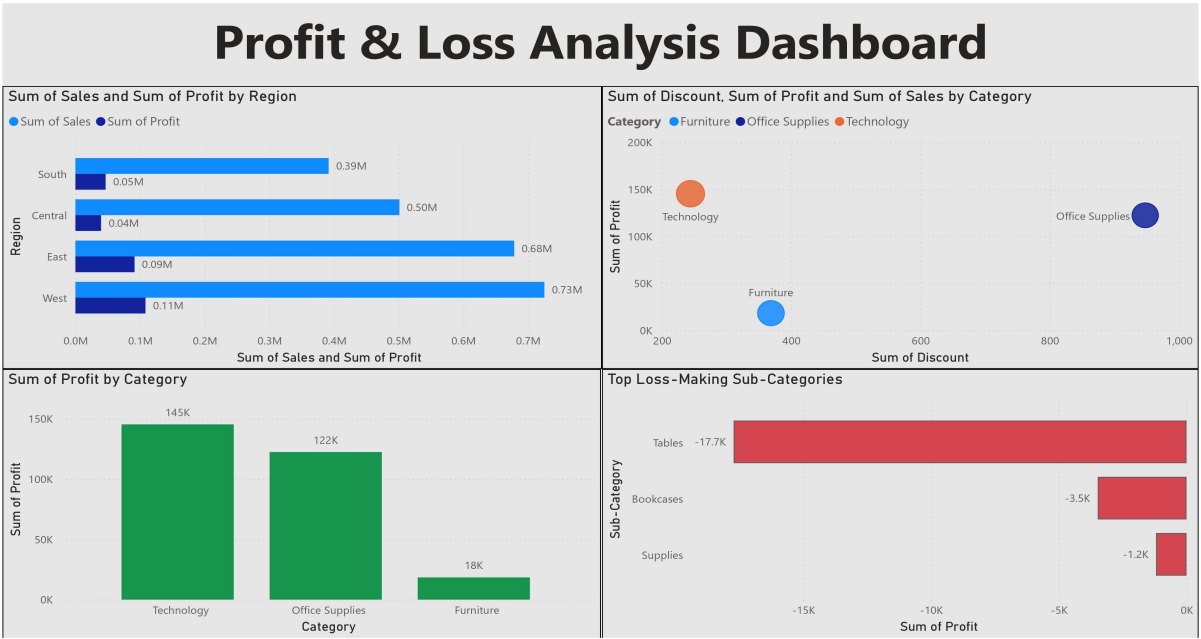


Figure 2: Profit & Loss Analysis — Page 2

4. Key Features

Sentiment Slicer	Filters all visuals dynamically by customer sentiment (Positive, Neutral, Negative) — no technical knowledge required.
KPI Cards	Instant overview of Total Sales, Total Profit, and Total Quantity at the top of the dashboard.
Map Visualization	Bing Map with bubble markers showing sales volume across cities and regions.
Scatter Plot	Reveals the correlation between Discount, Sales, and Profit at the product category level.
Conditional Formatting	Green = Profit, Red = Loss — applied consistently to make performance instantly readable.
Loss Sub-Category Chart	Ranks top loss-making sub-categories (e.g., Tables at -17.7K) to prioritize corrective action.
Segment Pie Chart	Breaks revenue down by Consumer, Corporate, and Home Office customer segments.
Regional Comparison	Side-by-side bars for Sales vs. Profit across South, Central, East, and West regions.

5. Tech Stack

Power BI Desktop	Primary platform for data modeling, dashboard design, and interactive visualization publishing.
Sample Superstore	Retail dataset with Sales, Profit, Discount, Region, Category, Sub-Category, and Segment fields.
DAX	Used to compute Total Sales, Total Profit, and Total Quantity measures powering the KPI cards.
Chart Types	Bar, Pie, Scatter Plot, Map (Bing), Funnel, and Grid — chosen to communicate each insight optimally.

6. Unique Points

- **Sentiment-Driven Analysis:** Connects customer feedback sentiment directly with sales and profit metrics — uncommon in standard BI dashboards.
- **Dual-Page Architecture:** Separates overview from deep analysis, reducing cognitive load and improving navigation.
- **Proactive Loss Identification:** The loss chart quantifies and ranks negative contributors, enabling targeted management action.
- **Fully Interactive & Accessible:** All visuals respond to slicers; no SQL or programming knowledge needed to explore data.
- **Clean Design with Color Coding:** Consistent green/red profit-loss scheme across all visuals ensures instant readability.

7. Future Improvements

- Real-Time Data Integration: Connect to live databases for automatic dashboard refresh without manual updates.
- AI/NLP Sentiment Analysis: Replace manual labels with an NLP model (NLTK or Azure AI) for automatic feedback classification.
- Expanded Filters: Add Region, Category, and Time Period (Year/Month/Quarter) slicers for deeper exploration.
- Predictive Analytics: Use Power BI forecasting or Azure ML to project future sales and profit trends.
- Mobile Optimization: Design a responsive mobile layout for executive access from smartphones and tablets.

8. Conclusion

This project demonstrates how data visualization bridges the gap between raw customer feedback and actionable business intelligence. By integrating sentiment analysis with sales and profitability metrics in an interactive Power BI dashboard, the solution enables decision-makers to quickly identify trends, diagnose problems, and act effectively — without needing technical expertise.

Key outcomes: clear profit/loss identification across regions and categories; deeper understanding of customer sentiment patterns; improved, data-driven decision-making; and a scalable foundation for future AI-powered and real-time enhancements.
